Every node in the tree is the root of some subtree.

(●) T ( ) F

评测结果 答案正确

得分 2 分

R1-2 分数 2 作者 何钦铭 单位 浙

If the five integers {1,2,3,4,5} are inserted into an initially empty binary search tree in any order starting with 3, the resulting trees will all have t same height.

(●) T ( ) F

评测结果 答案正确

得分 2 分

R1-3 分数 2 作者 陈超超 单位 浙江大学

Given a max-heap with unique keys, if $i < j$, then the element p in level i must be larger than q in level j.

( ) T (●) F

评测结果 答案正确

得分 2 分

R1-4 分数 2 作者 陈佳伟 单位 浙江大学

Suppose that each single node in a disjoint set initially has height 0, the height of the resulting disjoint-set tree formed by performing union-by-height on n nodes will not exceed $\lfloor \log_2 n \rfloor$.

(●) T ( ) F

评测结果 答案正确

得分 2 分

R1-5 分数 2 作者 何钦铭 单位 浙江大学

If a complete binary tree has $n+1$ nodes in the last level and n nodes in the second last level, then the total number of nodes in the tree must be $3n$.

(●) T ( ) F

评测结果 答案正确

得分 2 分

R1-6 分数 2 作者 DS课程组 单位 浙江大学

If N numbers are stored in a singly linked list in increasing order, then the average time complexity for binary search is $O(logN)$.

- T
- F

评测结果 答案正确

得分 2 分

R1-7 分数 2 作者 李松 单位 浙江大

$114514^N = O(N!)$

- T
- F

评测结果 答案正确

得分 2 分

R2-1 分数 3 作者 陈翔 单位 浙江大学

Suppose we have a circular queue implemented on an array `a[20]`. We store the position of the first queue element in `front`, and the next position of the last queue element in `rear`. What is the size of the circular queue when `front==7` and `rear==3`?

- A. 5
- B. 4
- C. 16
- D. 17

评测结果 答案正确

得分 3 分

R2-2 分数 3 作者 朱建科 单位 浙江大学

For an in-order threaded binary tree, if the post-order and in-order traversal sequences are FDEBGCA and FDBEACG, respectively. Which pair of nodes' right links are both threads?

- A. C and D.
- B. C and F.
- C. E and G.
- D. B and G.

评测结果 答案正确

得分 3 分

R2-3 分数 3 作者 陈佳伟 单位 浙江大学

The array representation of the disjoint sets is given by { 4, 5, 4, 5, -6, 9, -2, 7, -3, 9, 2 }. Keep in mind that the elements are numbered from 1 to 11, and the depth of the root is 0. After invoking `Union(Find(6), Find(8))`, `Union(Find(10), Find(3))` with union-by-size and path compression, what is the maximum depth of all the nodes?

- A. 4
- B. 1
- C. 3
- D. 2

评测结果 答案正确

得分 3 分

R2-4 分数 3

作者 陈越 单位 浙江大

A "full tree" of degree 3 (every non-leaf node has 3 children) has 61 nodes. Then its height h is at most ___. Note: $h = 0$ for a single node tree.

A. cannot be determined B. 21 C. 20 D. 19

评测结果 答案正确

得分 3 分

R2-5 分数 3

作者 陈超超 单位 浙江大学

Design an algorithm to determine whether the parentheses `()` in an expression are properly matched. Which data structure is the most suitable for this task?

A. Queue B. Sequentially stored linear list C. Heap D. Stack

评测结果 答案正确

得分 3 分

R2-6 分数 3

作者 陈超超 单位 浙江大学

It is known that a 3-heap is a heap whose nodes have 3 children. In a min-3-heap with 2025 elements, which one of the following indices is NOT a possible position for the maximum key?

A. 676 B. 1012 C. 1013 D. 675

评测结果 答案正确

得分 3 分

R2-7 分数 3

作者 朱建科

Consider the following operations performed on an initially empty Binary Search Tree (BST):
Insert: 50, 30, 70, 20, 40, 60, 80
Delete: 50 (replaced with its in-order successor)
Insert: 55
After these operations, which node will be the in-order predecessor of 60 in the final BST?

A. The in-order predecessor does not exist for 60.

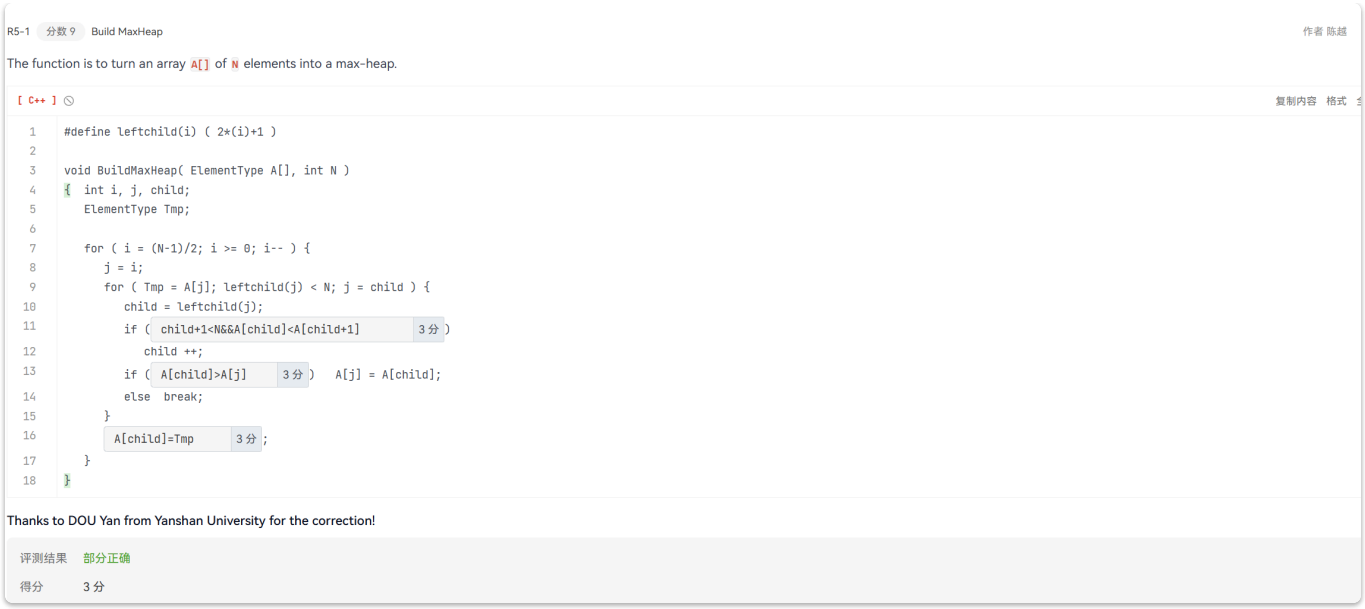
B. 40.

C. 55.

D. 30.

评测结果 答案正确

得分 3 分



函数题：

You are supposed to write a function of constructing a binary tree with given postorder and inorder sequences.

Format of function:

`Tree BuildTree( int inorder[], int postorder[], int N );`

where `Tree` is defined as the following:

```
typedef struct TreeNode *Tree;
struct TreeNode {
    int Element;
    Tree Left;
    Tree Right;
};
```

The inorder and postorder sequences are stored in `int inorder[]` and `int postorder[]`, respectively. The integer `N` is the number of nodes in the tree. The function `BuildTree` is supposed to return the pointer to the root of the constructed tree.

Sample program of judge:

```
#include <stdio.h>
#include <stdlib.h>


#define MAXN 10

typedef struct TreeNode *Tree;
```

```
struct TreeNode {
    int Element;
    Tree Left;
    Tree Right;
};

Tree BuildTree( int inorder[], int postorder[], int N );

void Inorder_output( Tree T ); /* details omitted */
void Postorder_output( Tree T ); /* details omitted */

int main()
{
    Tree T;
    int inorder[MAXN], postorder[MAXN], N, i;
    scanf("%d", &N);
    for (i=0; i<N; i++) scanf("%d", &inorder[i]);
    for (i=0; i<N; i++) scanf("%d", &postorder[i]);
    T = BuildTree(inorder, postorder, N);
    printf("Check:\n");
    Inorder_output(T); printf("\n");
    Postorder_output(T); printf("\n");
    return 0;
}
```

/ *Your function will be put here* /

Sample Input:

7

1 2 3 4 5 6 7

2 3 1 5 7 6 4

Sample Output:

Check:
1 2 3 4 5 6 7
2 3 1 5 7 6 4
个人答案：

```
Tree BuildTree( int inorder[], int postorder[], int N ){
    if(N==0){
        return NULL;
    }
```

```
    Tree newnode = malloc(sizeof(Tree));
    int root = postorder[N-1];
    newnode->Element=root;
    int k;
    for(int i = 0;i<N;i++){
        if(inorder[i]==root){
            k = i;
            break;
        }
    }
    newnode->Left=BuildTree(inorder,postorder,k);
    newnode->Right=BuildTree(inorder+k+1,postorder+k,N-k-1);
    return newnode;
}
```